Tutorial 9

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① Suppose that $c \leq d$ are points in $[a,b]$. If $\varphi : [a,b] \longrightarrow \mathbb{R}$ is given by $(\gamma > 0)$

$$\varphi(x) = \begin{cases} \gamma, & \text{if } x \in [c,d] \\ 0, & \text{elsewhere in } [a,b] \end{cases}$$

Prove that $\varphi \in \mathcal{R}[a,b]$

and $\displaystyle\int_a^b \varphi(x)\,dx = \alpha(d-c).$

(2) Let $\dot{P} = \{(I_i, t_i)\}_{i=1}^{n}$ be a tagged partition of $[a,b]$ and let $c_1 < c_2$.

(a) If $u \in I_k$ with $c_1 \leq t_k \leq c_2$ then show that

$$c_1 - \|\dot{P}\| \leq u \leq c_2 + \|\dot{P}\|.$$

(b) If $v \in [a,b]$ with

$$c_1 + \|\dot{P}\| \leq v \leq c_2 - \|\dot{P}\|$$

then the tag t_i of any subinterval I_i that contains v satisfies $t_i \in [c_1, c_2]$

(3) If $f \in \mathcal{R}[a,b]$ and if $\{\dot{P}_n\}$ is any sequence of tagged partitions of $[a,b]$ such that $\|\dot{P}_n\| \to 0$,

then show that

$$\int_a^b f(x)\,dx = \lim_n S(f;\dot{P}_n)$$

(4) Let

$$g(x) = \begin{cases} 0 & \text{if } x \in [0,1] \text{ is rational} \\ \frac{1}{x} & \text{if } x \in [0,1] \text{ is irrational.} \end{cases}$$

Explain why $g \notin \mathcal{R}[0,1]$.

However, show that there is a sequence $\{\dot{P}_n\}$ of tagged partitions of $[a,b]$ s.t

$\|\dot{P}_n\| \longrightarrow 0$ and $\lim_n S(f;\dot{P}_n)$ exists.

⑤ Let $f \in \mathcal{R}[a,b]$ and $c \in \mathbb{R}$.

Define $g : [a+c, b+c] \longrightarrow \mathbb{R}$

by $g(y) := f(y-c)$.

Show that $g \in \mathcal{R}[a+c, b+c]$

and that

$$\int_{a+c}^{b+c} g(x)\,dx = \int_a^b f(x)\,dx.$$

⑥ Is "Dirchlet function" Riemann integrable over $[0,1]$?

(7) Suppose $f : [a,b] \longrightarrow \mathbb{R}$ is a function s.t $f(x) = 0$ except for a finite number of points $c_1, c_2, c_3, \ldots, c_n$ in $[a,b]$.

Prove that $f \in R[a,b]$ and

$$\int_a^b f(x)\,dx = 0.$$

(8) If $g \in R[a,b]$ and if $f(x) = g(x)$ except for finite number of points in $[a,b]$ then show that $f \in R[a,b]$ and $\int_a^b f(x)\,dx = \int_a^b g(x)\,dx$

(9) If $f, g \in \mathcal{R}[a,b]$ and

$f(x) \leq g(x)$ for all $x \in [a,b]$

then $\displaystyle\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx$?

(10) Define $h : [0,1] \longrightarrow \mathbb{R}$ by

$$h(x) = \begin{cases} x+1 & \text{for } x \in [0,1] \text{ is rational} \\ 0 & \text{for } x \in [0,1] \text{ is irrational} \end{cases}$$

Show that $h \notin \mathcal{R}[0,1]$.

(11) Define $u : [0,1] \longrightarrow \mathbb{R}$ by

$$u(x) = \begin{cases} n & \text{if } x = \frac{1}{n}, n \in \mathbb{N} \\ 0 & \text{otherwise.} \end{cases}$$

Is u Riemann Integrable?

(12) Show that a step function $\phi : [a,b] \longrightarrow \mathbb{R}$ is Riemann integrable.

(13) If $S(f; \dot{P})$ is any Riemann Sum of $f : [a,b] \to \mathbb{R}$

Show that $\exists$ a step function $\varphi : [a,b] \longrightarrow \mathbb{R}$ s.t

$$\int_a^b \varphi(x)\,dx = S(f; \dot{P}).$$

(14) Suppose $f : [a,b] \longrightarrow \mathbb{R}$ is continuous and $f(x) \geq 0$ for all $x \in [a,b]$.

prove that if $\int_a^b f(x)\,dx = 0$

then $f(x) = 0$, $\forall x \in [a,b]$.

(15) If f, g are Continuous functions on $[a,b]$. and

$$\int_a^b f(x)\,dx = \int_a^b g(x)\,dx.$$

Prove that there is $c \in [a,b]$ s.t $f(c) = g(c)$.